CSAI 102: Digital Design Project Report

Design and Implementation of a 2-Digit Stopwatch

Mohammed Ayman
Youssuf Hosameldin
Anas Abdelmoghny
Amar Elsagey
Rewan

December 22, 2024

Contents

1 Introduction **2**

2 Circuit Design Diagram **2**

3 Component List **3**

4 Design Description **3**
4.1 Component Selection and Functionality 3
4.2 Avoiding Floating Point Inputs 4

5 Cost of the Circuit **5**

6 Conclusion **5**
6.1 Component Datasheets 6
6.2 References 6

1 Introduction

This report details the design and implementation of a digital stopwatch capable of counting from **00** to **99**. The stopwatch leverages integrated circuits (ICs) such as the `CD4026` for counting and display, a `555 Timer` for clock generation, and standard components like push buttons and jumper wires to facilitate user interaction and circuit connectivity. The project emphasizes the use of readily available components to create a cost-effective and reliable timing device.

2 Circuit Design Diagram

Figure 1 illustrates the circuit design of the 2-digit stopwatch created using Logisim (or your chosen simulator).

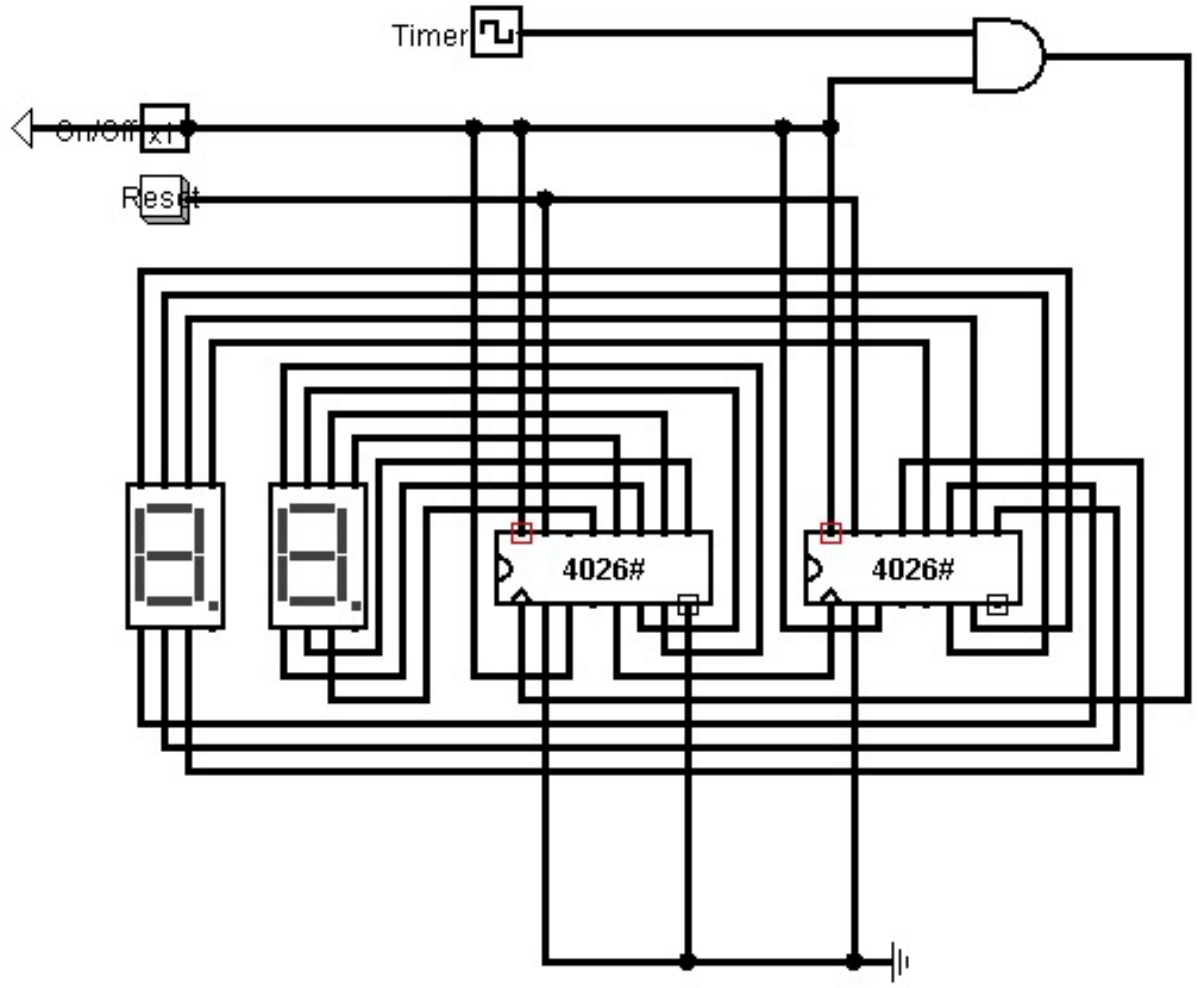


Figure 1: Circuit Diagram of the 2-Digit Stopwatch

Description of the Circuit Diagram:

- **Clock Generation:** A `555 Timer` is configured in astable mode to generate a consistent clock signal.
- **Counter and Display Mechanism:** Two `CD4026` ICs are used to count and drive two 7-segment displays, enabling the stopwatch to count from **00** to **99**.

- **Control Mechanisms:** An on/off switch manages the start/stop functionality, while push-button switches allow for resetting the counter to zero.
- **Connectivity:** Jumper wires facilitate connections between components on a breadboard, ensuring proper signal flow and functionality.

3 Component List

Component	Part Number	Quantity	Description
555 Timer	NE555	1	Configured as astable oscillator for clock generation
BCD Counter & Decoder	CD4026	2	Combines BCD counting and 7-segment decoding
7-Segment Display	Common Anode	2	Displays numerical count
Jumper Wires	-	50	For connecting components on the breadboard
Push-Button Switch	SPST	2	One for start/stop, one for reset
Resistors	Various	-	For current limiting and pull-ups
Capacitors	Various	-	For timing in the 555 oscillator
Breadboard	-	1	For assembling the circuit
Power Supply	5V DC	1	Provides necessary operating voltage

Note: Additional passive components (resistors, capacitors) are included as needed for proper circuit functionality.

4 Design Description

4.1 Component Selection and Functionality

1. **555 Timer (NE555):**
   - **Reason for Selection:** The NE555 timer is a versatile and cost-effective component widely used for generating precise clock signals in various digital circuits.
   - **Function:** Configured in astable mode, the 555 timer produces a steady clock pulse that drives the CD4026 counters at a defined frequency, ensuring accurate time measurement.

2. **BCD Counter & Decoder (CD4026):**
    - **Reason for Selection:** The CD4026 is a dual-function IC that integrates both BCD counting and 7-segment decoding, simplifying the design by reducing the number of required components.
    - **Function:** Each CD4026 handles one digit of the stopwatch, counting from 0 to 9 and driving the corresponding 7-segment display. Cascading two CD4026 ICs allows the system to count from **00** to **99** seamlessly.
3. **7-Segment Displays:**
    - **Reason for Selection:** Common anode 7-segment displays are easy to interface with the CD4026 decoder and provide clear numerical output.
    - **Function:** Visually display the count from **00** to **99**, allowing users to read the stopwatch time accurately.
4. **Push-Button Switches:**
    - **Reason for Selection:** Simple SPST (Single Pole Single Throw) switches provide reliable user control over the stopwatch operations.
    - **Function:** One push-button switch serves as the start/stop control, enabling users to begin or halt the counting process. The second push-button switch functions as a reset, initializing the counter back to zero when pressed.
5. **Jumper Wires:**
    - **Reason for Selection:** Essential for creating connections between components on the breadboard, ensuring signal integrity and proper functionality.
    - **Function:** Facilitate the routing of electrical signals between the 555 timer, CD4026 ICs, 7-segment displays, and switches.
6. **Passive Components (Resistors, Capacitors):**
    - **Reason for Selection:** Necessary for stabilizing the circuit, limiting current to the LEDs in the 7-segment displays, and setting the timing parameters for the 555 oscillator.
    - **Function:** Ensure reliable operation and protect active components from electrical overstress.

4.2 Avoiding Floating Point Inputs

Floating inputs in digital circuits can lead to unpredictable behavior, increased power consumption, and susceptibility to noise. To mitigate these issues in the stopwatch design:

1. **Pull-Up/Pull-Down Resistors:**
    - **Implementation:** All unused inputs on integrated circuits (e.g., CD4026) are connected to a defined logic level using pull-up or pull-down resistors. This prevents inputs from floating by maintaining them at a stable high or low voltage level.

2. **Proper Wiring:**
    - **Implementation:** All input pins, especially those controlling the counting and resetting functions of the CD4026, are tied to either the power supply (Vcc) or ground (GND) through appropriate resistors, ensuring they are not left floating.
3. **Debouncing Switches:**
    - **Implementation:** Mechanical switches can introduce noise when toggled. Incorporating debounce circuits (using resistors and capacitors) ensures that switches provide clean, stable signals, preventing unintended floating states.
4. **Simulation Validation:**
    - **Implementation:** Utilizing simulation tools like Logisim helps identify and rectify any floating inputs before physical implementation, ensuring robust circuit behavior.

5 Cost of the Circuit

Component	**Quantity**	**Unit Price (LE)**	**Total Cost (LE)**
NE555 Timer	1	4	4
CD4026 IC	2	20	40
7-Segment Display	2	5	10
Jumper Wires	50	1	50
Push-Button Switch	2	2	4
Resistors	Various	-	5
Capacitors	Various	-	5
Breadboard	1	10	10
Power Supply (5V DC)	1	15	15
Total Estimated Cost			**143**

Note: Prices are based on the components provided by the user. Additional passive components like resistors and capacitors are estimated and may vary based on specific requirements.

6 Conclusion

The designed stopwatch successfully utilizes integrated circuits and standard components to achieve a functional counting mechanism from **00** to **99**. By selecting the `CD4026` ICs, the design simplifies the counting and decoding process, reducing the number of required components while maintaining reliability and accuracy. The `555 Timer` provides a stable clock source, ensuring precise time measurement. Thoughtful design considerations, such as avoiding floating inputs, enhance the circuit's robustness and prevent potential malfunctions. The project demonstrates effective digital design principles and offers a cost-efficient solution for basic time measurement applications.

6.1 Component Datasheets

- **CD4026 IC:** CD4026B Datasheet
- **NE555 Timer:** NE555 Datasheet

6.2 References

1. **555 Timer Circuits:** Electronics Tutorials - 555 Timer
2. **CD4026 Applications:** CD4026 Datasheet and Application Notes
3. **Avoiding Floating Inputs in Digital Circuits:** All About Circuits - Pull-Up and Pull-Down Resistors Explained
4. **Debouncing Mechanical Switches:** Electronics Tutorials - Switch Debouncing